Rekia Fadla

Final-Year Computer Systems Engineering Student at ESI-SBA | Seeking Internship

r.fadla@esi-sba.dz | roukaiafadla | fadlarekia | Portfolio | Lghouat, Algeria

SUMMARY

Final-year Computer Systems Engineering student at ESI-SBA. Experienced in building end-to-end IoT systems—from hardware wiring and microcontroller firmware to deployed backends, live dashboards, and applied machine learning pipelines. Additional strong foundations in full-stack development and network security. Seeking a research internship in embedded systems, edge AI, or applied ML.

EDUCATION

Higher School of Computer Science (ESI-SBA), Sidi Bel Abbès, Algeria *2022 – 2027*
Engineering Degree in Computer Science — Systems Engineering (ISI) — Expected 2027.
Baccalaureate in Mathematics — 2022 — Graduated with honors, 17.25/20.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript (Node.js), SQL, PL/SQL, R, MATLAB
Embedded & IoT: ESP32, Arduino, Raspberry Pi, ARM, SPI, I^2C, UART, MQTT, embedded Linux (buildroot).
Networking: TCP/IP, VLANs, dynamic routing (RIP, OSPF, BGP), QoS, IPv6, SNMP, Wireshark, Packet Tracer.
AI & Data: scikit-learn, OpenCV, YOLOv12 (ONNX), EasyOCR, NumPy, Weka, PCA, TensorFlow/Keras.
Backend & Tools: Node.js, Express, Flask, REST, WebSocket, MongoDB, SQL, Git, Linux, Render, Vercel.

SELECTED PROJECTS

PlantCare — End-to-end IoT irrigation and telemetry system *Solo*
github.com/roukaiafadla/plantcare • live dashboard
- Built a complete IoT system: ESP32 firmware samples soil, temperature, and humidity every 5 s and publishes JSON over MQTT.
- Engineered local auto-watering logic on the ESP32 for fail-safe offline operation; built the Node.js/MongoDB backend and deployed the live WebSocket dashboard.

Smart Parking — RFID and license-plate access control *Team of 4*
github.com/roukaiafadla/smart-parking • supervised by Prof. A. Rahmoun
- Co-built a distributed access-control system integrating Raspberry Pi 4, Arduino, Flask, and MongoDB with fail-safe barrier behavior.
- Led hardware integration (RFID, Pi Camera, sensors) and built the admin dashboard; integrated a two-stage computer vision pipeline (YOLOv12 ONNX + EasyOCR).

ml-lab — Machine learning algorithms from scratch with NumPy *Solo*
github.com/roukaiafadla/ml-lab
- Implemented linear and logistic regression from scratch (vectorized gradient descent, analytic gradients verified against finite differences).
- Achieved test RMSE of 0.7457 vs 0.7456 for scikit-learn; built reproducible experiments, unit tests (pytest), and CI (GitHub Actions).

Wassit — AI-assisted hyperlocal services marketplace *Solo*
github.com/roukaiafadla/wassit
- Full-stack platform using React, Node.js, and Socket.io for real-time service broadcasts.
- Integrated Google Gemini API to parse free-text job descriptions and automatically suggest category, urgency, and pricing.

Wanderly & Shaghaf — Full-stack web applications *Solo*
wanderly live • code • shaghaf live • code
- Developed and deployed production-ready MERN and Node.js applications featuring JWT authentication, Cloudinary uploads, and CI-tested APIs.

PROFESSIONAL EXPERIENCE

Network Engineering Intern — Algérie Télécom, Laghouat *September 2025*
- Analyzed core network infrastructures, fiber-optic transmission centers, and GPON/FTTH access architectures at the national operator's regional headquarters.
- Explored enterprise data systems (Oracle Finance, HR Access) and routing hardware (Cisco, Huawei); practiced VLAN and RIP configuration via Cisco Packet Tracer.

LANGUAGES

Arabic (Native) | French (B2) | English (B2).